

[详解] (1) 过 $F(p/2, 0)$ 、倾斜角 $\pi/4$ 的直线为 $y = x - p/2$. 代入 $y^2 = 2px$: $(x - p/2)^2 = 2px$, 即 $x^2 - 3px + p^2/4 = 0$ ($\Delta = 9p^2 - p^2 = 8p^2 > 0$). 设 $A(x_A, y_A)$ 、 $B(x_B, y_B)$, 则 $x_A + x_B = 3p$, $x_A x_B = p^2/4$. 由定义 (焦半径),

$$|AB| = |AF| + |BF| = (x_A + p/2) + (x_B + p/2) = x_A + x_B + p = 4p.$$

(2) 该弦所在直线与 x 轴交于 F , 两端点 A 、 B 分居 x 轴两侧, 纵坐标一正一负, 其积为负:

$$y_A y_B = (x_A - p/2)(x_B - p/2) = x_A x_B - (p/2)(x_A + x_B) + p^2/4 = p^2/4 - 3p^2/2 + p^2/4 = -p^2 \quad (\text{与焦点弦结论 } y_1 y_2 = -p^2 \text{ 相符, 此即符号判据}).$$

$$\text{于是 } x_A x_B + y_A y_B = p^2/4 - p^2 = -3p^2/4. \text{ 又 } |OA|^2 = x_A^2 + y_A^2 = x_A^2 + 2px_A = x_A(x_A + 2p), \text{ 同理 } |OB|^2 = x_B(x_B + 2p).$$

$$\text{故 } |OA| \cdot |OB| \text{ 等于 } \sqrt{x_A x_B} \text{ 与 } \sqrt{x_A x_B + 2p(x_A + x_B) + 4p^2} \text{ 之积, 代入韦达读数得 } (p/2) \cdot \sqrt{p^2/4 + 6p^2 + 4p^2} = (p/2) \cdot (\sqrt{41}/2)p = \sqrt{41}p^2/4.$$

$$\text{在 } \triangle AOB \text{ 中由余弦定理, } \cos \angle AOB = (|OA|^2 + |OB|^2 - |AB|^2) / (2|OA| \cdot |OB|).$$

$$\text{而 } |OA|^2 + |OB|^2 - |AB|^2 = (x_A - x_B)^2 + (y_A - y_B)^2 = 2(x_A x_B + y_A y_B) = -3p^2/2,$$

$$\text{故 } \cos \angle AOB = (-3p^2/4) / (\sqrt{41}p^2/4) = -3/\sqrt{41} = -3\sqrt{41}/41, p^2 \text{ 全部约去, 与 } p \text{ 无关.}$$

[详解] $M(x, y)$ 在以 $B(6, 0)$ 为圆心、2 为半径的圆上:

$$(x-6)^2 + y^2 = 4; N(x_0, y_0) \text{ 满足 } y_0^2 = 8x_0 \quad (x_0 \geq 0). \text{ 在圆约束下加权配凑:}$$

$$(1/4)|MA|^2 = (1/4)[(x-2)^2 + y^2] = (1/4)[(x-2)^2 + y^2] + (3/4)[(x-6)^2 + y^2 - 4] = (x-5)^2 + y^2 \quad (\text{所加项在圆上为 } 0), \text{ 即 } (1/2)|MA| = |MC|, \text{ 其中 } C(5, 0).$$

$$\text{于是 } |MN| + (1/2)|MA| = |MN| + |MC| \geq |CN|.$$

$$\text{又 } |CN|^2 = (x_0 - 5)^2 + y_0^2 = (x_0 - 5)^2 + 8x_0 = (x_0 - 1)^2 + 24 \geq 24, \text{ 故 } |CN| \geq 2\sqrt{6}, \text{ 当且仅当 } x_0 = 1 \quad (N(1, \pm 2\sqrt{2})) \text{ 且 } C、M、N \text{ 三点共线} \quad (M \text{ 为线段 } CN \text{ 与圆的交点}) \text{ 时取等——取等可行: } C(5, 0) \text{ 在圆内} \quad (|CB| = 1 < 2) \text{ 而 } N \text{ 在圆外} \quad (|NB|^2 = (x_0 - 6)^2 + 8x_0 = (x_0 - 2)^2 + 32 > 4 \text{ 恒成立}), \text{ 线段 } CN \text{ 必与圆相交. 最小值 } 2\sqrt{6}, \text{ 选 D.}$$